Skills Summary

Measurement Conversion, Volume, and Angle Measure

Use this reference while completing your worksheet or when studying.

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1. Converting units (unit-conversion)

Rule: A *bigger* unit turned into a *smaller* one gives *more* of them, so **multiply**. Going the other way, **divide**.

1 ft = 12 in   1 yd = 3 ft   1 m = 100 cm   1 km = 1000 m
1 kg = 1000 g   1 L = 1000 mL   1 h = 60 min   1 min = 60 s

Mixed measures (2 kg 250 g) convert the big part, then **add** the leftover.

Example: A ribbon is 5 m long. How many centimetres is that?
Step 1. Metres are bigger than centimetres, so the number of centimetres must be larger — that means multiply by the 100 in 1 m = 100 cm.
Step 2. $5 \cdot 100 = 500$ The ribbon is **500** cm long.

Try it: A bench is 7 ft long. How many inches is that?

check: **84** in

(!) Watch out: Answer in the unit the question *asks* for, not the unit the numbers came in — and never multiply two edges that are still in different units.

2. Volume of a rectangular prism (prism-volume)

Formula: V = length · width · height, which is the same as (area of the base) · height.

Volume is measured in **cubic** units (cm^3, ft^3) because you are counting unit cubes that fill the box.

Missing edge? Divide: height = $V \div$ (base area).

Example: A box is 7 cm long, 3 cm wide, and 2 cm high. Find its volume.
Step 1. All three edges are given in the same unit and the box is a prism, so multiply the three edges — no conversion needed.
Step 2. Bottom layer: $7 \cdot 3 = 21$ cubes. Two layers: $21 \cdot 2 = 42$. $V = \mathbf{42}\ \text{cm}^3$.

Try it: A crate is 9 cm by 2 cm by 4 cm. Find its volume.

check: **72** cubic cm

3. Angle measure (angle-measure)

Rule: Angles that sit side by side **add**: if ray BD is inside $\angle ABC$, then $\angle ABD + \angle DBC = \angle ABC$.

Whole angles worth knowing: a right angle is $90°$, a straight angle is $180°$, a full turn is $360°$.

Missing part? part = whole − known part.

Example: A right angle is split into two parts. One part is $52°$. Find the other.
Step 1. The two parts fill a right angle, so their whole is $90°$ and the missing part is found by subtracting, not by adding.
Step 2. $90 - 52 = 38$ The other part is **38** degrees.

Try it: Two angles together form a straight angle. One measures $118°$. Find the other.

check: **62** degrees

4. Putting it together (multistep-synthesis)

Plan for a multistep problem:

(1) Underline the unit the question asks for. **(2)** Make every given measure match before you multiply.
(3) Do the volume or angle work. **(4)** Convert once, at the end: $1000 \text{ cm}^3 = 1$ L.

Example: A tank is 30 cm by 20 cm by 25 cm. How many litres does it hold?
Step 1. The edges are in centimetres but the answer must be in litres, so find the volume first and convert at the very end.
Step 2. $30 \cdot 20 \cdot 25 = 15000 \text{ cm}^3$, then $15000 \div 1000 = 15$. The tank holds **15** L.

Try it: A tank is 50 cm by 20 cm by 40 cm. How many litres does it hold?

check: **40 L**